FORM TP 2017046

TEST CODE **01207032**

MAY/JUNE 2017

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

BIOLOGY

Paper 032 – General Proficiency

Alternative to School-Based Assessment

2 hours 10 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. You are advised to take some time to read through the paper and plan your answers.
5. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
6. **If you use the extra page(s), you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. You are required to conduct an experiment using the following materials and procedure.

Materials

- 250 ml beaker containing 100 ml of water
- Filter paper marked with concentric circles as shown in Figure 1
- Long forceps
- Crystals of coloured substance
- Watch glass
- Stopwatch

Procedure

(i) Before starting the experiment, read through the procedure carefully, then construct a suitable table on page 5, to record the data you will collect.

(ii) Pour 100 ml of water into the beaker.

(iii) Place the beaker of water on the middle of the filter paper and let it stand until there is no movement of the water. **See Figure 1.**

(iv) Using a long pair of forceps, carefully place a crystal of the coloured substance on the bottom of the beaker of water, as close to the centre of the concentric rings as possible. Note the time that the crystal is placed.

(v) Record the time (in seconds) that it takes for the colour to spread to meet the edge of each concentric circle.

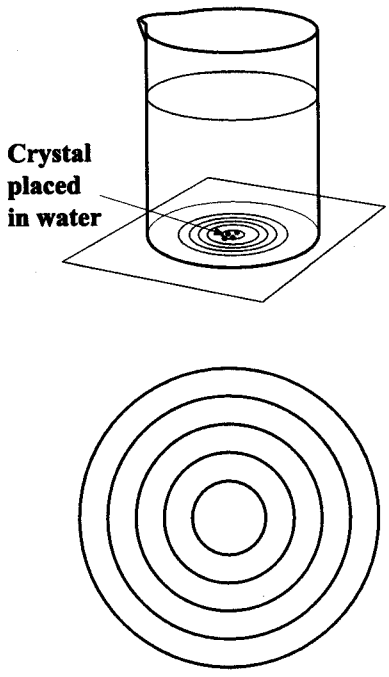

Figure 1. Beaker and filter paper with concentric circles

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(a) (i) Present the results in the table you constructed.

(5 marks)

(ii) Suggest how the experiment could be modified to investigate the effect of temperature on the spread of the colour through the water.

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(3 marks)

(iii) Explain the process by which the colour spreads through the water.

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(2 marks)

(iv) Explain how this process is important in plants and animals. Use ONE example for plants and ONE example for animals.

Plants .....

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Animals .....

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(4 marks)

(b) In another experiment, slices are cut from the cucumber shown in Figure 2.

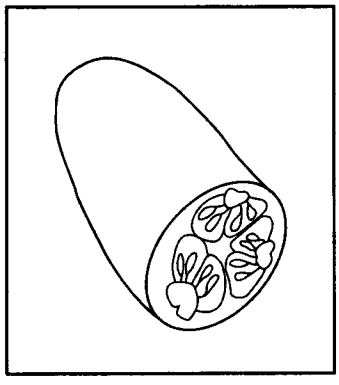

Figure 2. A cucumber cut in half

(i) In the box below, make a large, clearly labelled drawing to show a **cross section** of the cucumber, twice the size shown in Figure 2.

(5 marks)

(ii) Six slices of the cucumber are first measured. Three slices are then placed for one hour in each of two solutions:

Solution A – concentrated salt solution
Solution B – distilled water

Describe the expected appearance of the slices in EACH solution after one hour.

Solution A .....

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Solution B .....

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(2 marks)

(iii) Explain the process that may have contributed to the change in the appearance of the cucumber slices.

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(3 marks)

Total 24 marks

2. Figure 3 shows the organisms seen by students on a field trip to a wetland ecosystem.

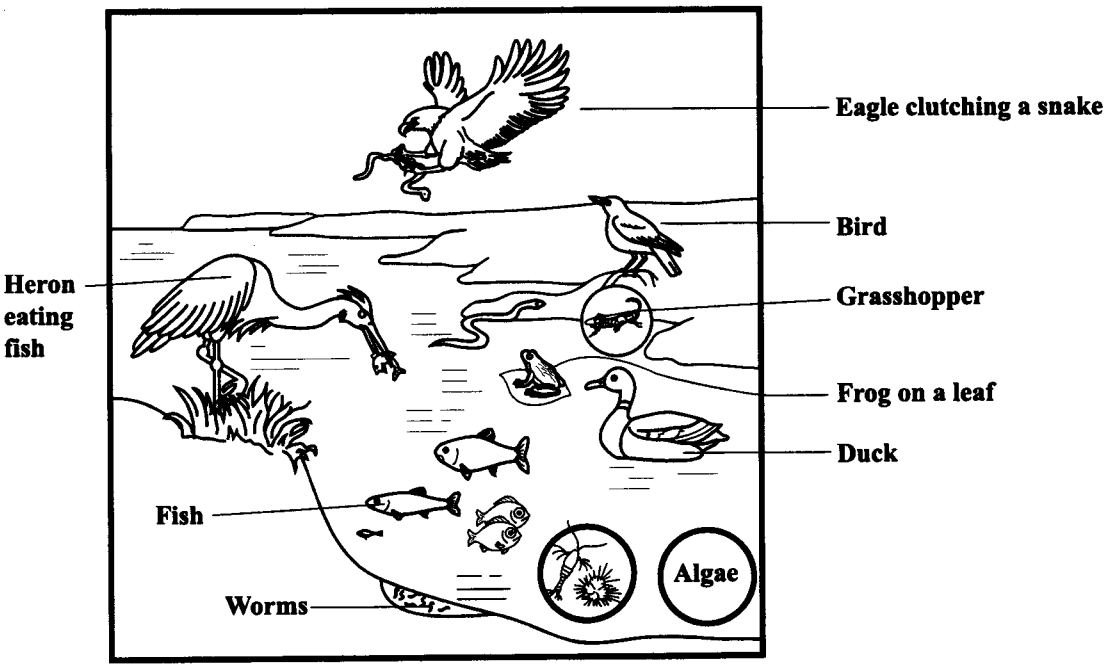

Figure 3. Organisms in a wetland ecosystem

(a) In the space below, construct a food web to show the feeding relationships between six organisms in Figure 3.

(4 marks)

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(b) Suggest TWO adaptations of the eagle that allow it to carry out its role, and state how EACH adaptation functions.

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(2 marks)

(c) (i) Describe ONE method that could be used to estimate the **change in the distribution** of a plant species from the water's edge to 100 m inland (away from the water's edge), and another method to estimate the **density** of the plant species.

Change in distribution .....

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Density .....

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(5 marks)

(ii) In the box below, draw a table that could be used to record the data for species density and state a title for the table.

(4 marks)

(d) Identify TWO methods for collecting aquatic animals.

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(2 marks)

(e) The students also collect soil samples from the shallow part of the pond. They take these samples back to the laboratory where they have access to the apparatus and materials shown in Figure 4.

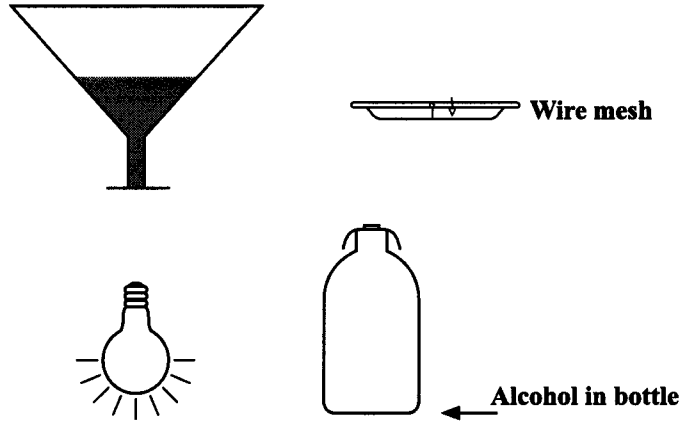

Figure 4. Apparatus and materials

Draw an annotated diagram in the box below to show how the apparatus and materials could be used to extract organisms from the soil the students collected.

(4 marks)

Total 21 marks

3. Figure 5 shows the sequence of events in a typical spinal reflex arc.

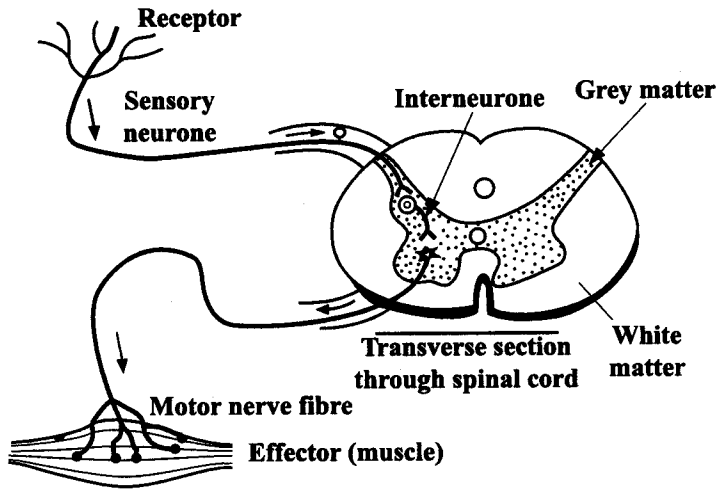

Figure 5. Sequence of events in a typical spinal reflex arc

(a) Describe a procedure that is used to test a person's knee jerk reflex.

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(3 marks)

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(b) A researcher conducts an experiment to test the hypothesis, 'athletes have a faster reaction time than non-athletes'. The data from this experiment are shown in Table 1.

TABLE 1: REACTION TIME OF ATHLETES AND NON-ATHLETES

Athlete	**1**	**2**	**3**	**4**	**5**
Reaction time	0.2	0.4	0.5	0.4	0.6
Non-athlete	**1**	**2**	**3**	**4**	**5**
Reaction time	0.5	1.0	1.1	1.2	1.0

(i) On the grid provided on page 15, draw two line graphs of the data in Table1.

(7 marks)

(ii) State ONE conclusion that can be drawn from the results of this experiment.

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(1 mark)

(iii) Describe ONE possible source of error in this experiment.

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(1 mark)

(iv) Identify THREE variables that should be controlled in this experiment.

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(3 marks)

Total 15 marks

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.